

Two-Step Pyrolysis of Spirogyra for Fuels Using Cement Catalytic

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Abstract Two-step catalytic pyrolysis of spirogyra, a filamentous alga commonly found in freshwater ditches and ponds, was investigated in a batch pyrolyser under static conditions. The effects of temperature, quantity of catalyst and retention time on properties of the pyrolysis products and yields of the oil, gases and solid residue were investigated. The liquid product obtained by catalytic pyrolysis of the powdered spirogyra was separated into three major fractions by distillation; i.e. a small quantity of flammable liquid, water and a brownish paste-like organic fraction. The paste was converted into a highly flammable gas and oil on catalytic pyrolysis, using cement as catalyst. This oil was analyzed and separated using GC-FID and GC-MS. It was found that the two-step catalytic pyrolysis of spirogyra has potential due to its yield, which had good liquid product density and low oxygen content.

Keywords Algae · Biomass · Two-step pyrolysis · Cement catalyst · Static batch reactor

Introduction

Aquatic plants and algae are important sources of biomass [1]. In comparison to terrestrial plants, algae is responsible for conversion of greater quantities of solar energy and fixation of more carbon dioxide [1, 2]. Studies on conversion of algae into fuel products using various technologies were reported by several workers in the last 10 years [3, 4]. Through pyrolysis, fuels in the forms of gases and oil products could be produced, these products contain large quantity of water [5], which may lead to increase in oxygen contents and most of these compounds are highly oxygenated. Consequently, the presence of large quantity of water and oxygenated compounds reduce the calorific value, increase the chance for more reactions, accelerates corrosion and cause problems in ignition [2].

Literature reports indicate that the oxygen content of the pyrolysates may be reduced through the use of various catalysts, such as zeolites, metal-substituted zeolites, platinum-supported catalyst, impregnated-salt catalysts and some free treatments [6]. In the present work spirogyra a big c was selected as biomass due to its easier availability and abundance. Another reason was the availability of a big chunk of literature about this fresh water filamentous alga. It has nearly zero lignin [7] and is considered as the best candidate for biomass conversion into oil and gas via pyrolysis technology. It is expected that the absence of lignin may help to obtain greater quantities of bio oil and gas at lower temperature. Just like other algae, it is also responsible for the fixation of greater quantities of the solar energy and is grown in water bodies as compared to

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terrestrial plants growing on fertile land, which can harm the production of agricultural-based food. It may also contain oily matter which is responsible for the increase in production of the oil and gas yield [7, 8]. Spirogyra has also been studied for generation of bio-char [8]. Its conversion into oil and gas products is considered as responsible for the fixation of large quantities of carbon dioxide into biomass.

The present work is aimed to study thermo-chemical conversion of spirogyra into fuel oil and gas by using catalytic pyrolysis. The work is also aimed to utilize a low cost catalyst for the biomass conversion process to eventually reduce the amount of water and oxygenated organic compounds, which are undesired in the liquid products [9].

Materials and Methods

Plant Material

Spirogyra is the name of a filamentous genus of algae. In the present work, Spirogyra link was used as biomass, which was selected on the basis of its abundance in Pakistan. The spirogyra used in this study was collected in October from a stream of Adina in the district of Swabi, which is located in the Province of Khyber Pakhtunkhwa. The collected biomass was initially washed with distilled water to remove mud and sludge like material. The alga was washed again with distilled water for the removal of water soluble mineral matter. The clean algal masses were then dried in a fume hood for 24 h. The dried alga was powdered using an indigenously prepared grinder machine and sieved using a 400 μm mesh. This work was mainly focused on the conversion of algae into the fuel products and the analysis of the resulting fuel. As the proximate and element analyses on the algae have already been reported by many researchers [10, 11], therefore were not considered in the presented work.

Pyrolysis and Reactor System

Finely powdered spirogyra was thoroughly mixed with the cement catalyst at a ratio according to requirements of the experiment. The mixture of spirogyra and catalyst was loaded into an indigenously prepared batch-type cylindrical stainless steel pyrolyser or reactor, as schematically shown in Fig. 1. The reactor had a lid which had a side tube for discharging vapors. The length of the side tube was 3.81 cm, internal diameter of the reactor was 6.7 cm and its height was 13.97 cm. A custom-made electric furnace was used as source of heating which was properly equipped with a thermocouple and a temperature controller system to set the temperature accordingly. A K type custom

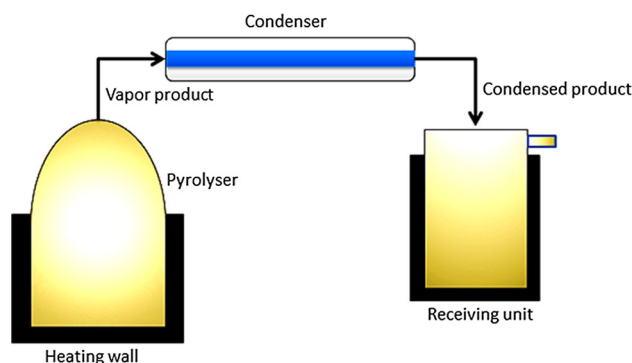


Fig. 1 Schematic of the reaction assembly

thermocouple of the fluke was used for sensing the temperature which can measure temperature to the extent of $\pm 0.4\%$ T. This was adjusted in a steel covering in the furnace in closure contact to pyrolyser for ensuring minimum temperature lag. The temperature for pyrolysis was explored in the range of 150–400 $^{\circ}\text{C}$. The side tube of the pyrolyser was connected to a condenser using Teflon tube. The products of pyrolysis were collected in a receiving vessel after condensation. This liquid product was concentrated through evaporation of water by distillation. The resulting sticky material was then pyrolysed after mixing with the catalyst at the same optimum ratio.

Analysis of the Product Oil

The oil obtained by second stage pyrolysis was analyzed using gas chromatography with flame ionization detector (GC FID) to optimize conditions for separation and quantitative analysis. The optimum conditions for the oil separation were analyzed using gas chromatography-mass spectrometry (GC-MS). The GC-MS was operated under split mode, while for GC FID it was run under splitless mode. Gas chromatographic analysis for the oil was carried out using its *n*-hexane solution on a 600 H Jeol GC-MS. A fused capillary column HP.5 of 30 m in length and 0.32 mm in internal diameter with a film thickness of 0.25 μm was used for separation of the components of the oil solution. The column contained polydimethylsiloxane (PDMS) as the stationary. Helium at a flow rate of 1.8 mL/min was used as carrier gas. An amount of 1 μL of the 0.1 % *n*-hexane solution of the oil was injected at a split ratio of 1/35 and injector temperature of 250 $^{\circ}\text{C}$. The column was isothermal at 60 $^{\circ}\text{C}$ for 2 min, after which the temperature was programmed to rise at a rate of 5 $^{\circ}\text{C}$ per minute until it reached 250 $^{\circ}\text{C}$. The temperatures of interface and detector were adjusted according to the temperature of injector. These conditions of gas chromatographic and mass spectrometric analyses of oil were investigated using the less expensive GC FID analysis. This was carried out by using Agilent 7890A gas

chromatograph with the same approach except for the use of nitrogen instead of helium as carrier gas.

Results and Discussions

Theoretical Basis of the Catalytic Pyrolysis

The products of pyrolysis of biomass contain large quantity of water which is mainly produced during the de-polymerization and decomposition of the cellulosic and ligno-cellulosic material of the biomass. However, spirogyra does not contain lignin [7, 8] and it is the decomposition of cellulosic matter which is responsible for the formation of liquid and gaseous products. The present work is focused on lowering the water content in liquid product of the pyrolysis. It is also aimed to reduce the oxygen content of the organic compounds. It was reported in our previous work [12] that cement catalyst could reduce the oxygen content of the organic compounds and could also reduce the formation of water either by trapping oxygen or by stabilizing the hydrogen evolved during the cracking process. Both options were feasible due to molecular holes of the cement which would also promote conversion of the lipid content of spirogyra into hydrocarbon oil. We have never worked on direct pyrolysis of spirogyra, however employed cement as catalyst. The oxygen contents of bio oil of direct pyrolysis of spirogyra were assumed in the range of 30–40 % based on the literature reports from the pyrolysis of various biomasses including algae [13, 14].

Optimum Pyrolysis Temperature

Catalytic pyrolysis is a temperature dependent reaction. Temperature may change both the rate of reaction and the nature of products of reaction [15]. The temperature may also change the behavior of catalyst due to changes in the vibrational energies of the active sites [16]. The activity of the active sites of catalyst may also change with changes in pressure [17]. It was observed that large quantities of gases would generate high pressure inside the pyrolyser during the process, and this pressure would increase with temperature. Changes in the pyrolysis temperature could also change the nature of products and the yield of pyrolysis, as reported by [7, 18]. In the present work, the effect of temperature on the nature and yield of the products of pyrolysis was investigated in the range of 150–400 °C by using 20 wt.% of the cement catalyst. The reaction duration was 1 h for all the experiments and each of the experiment was conducted in triplicate.

The experiment results are presented in Table 1. It can be seen from the results that the percent mass of the solid char decreases with increase in temperature, which results

in a proportional increase in the amount of liquid and gaseous product. This trend is in agreement with those reported in the literature [19]. Further increase in temperature results in increase in the density of the liquid product, and this implies reduction in water content. It can be seen from the results that the amount of liquid product and density are same at both 350 and 400 °C; however due to the small quantity of the resulting residue, the higher temperature (400 °C) was selected as the optimum temperature for further work [20].

Optimization of Pyrolysis Duration

The duration of reaction is an important parameter for almost all pyrolysis processes. The time taken for a reaction may have implication on cost, and also on the nature and relative amount of the product for an operation at a constant temperature. The optimum duration for the catalytic pyrolysis in the present work was investigated in the range of between 15 and 90 min. The experiments were conducted by heating 20 wt.% of the cement catalyst with fine powder of spirogyra at 400 °C in a pre-heated furnace. Each of the tests was conducted in triplicate to obtain the average values of products as shown in Table 2. Clearly, the table indicates that the amount of residue remains almost same for reaction duration of 60 min or longer and hence was selected as the optimum duration for onward tests.

Effect of Quantity of Catalyst

Catalyst may alter both the nature and yield of the pyrolysis products and may also change the rate of reaction [15, 19]. These variations in the rate of reaction, yields and nature of products vary according to the nature of feed stock and catalyst [21]. Variation in the relative amount of catalyst may also have a profound effect on the pyrolysis product [22]. This is very important in case of static system and reactions involving heterogeneous systems. The relative amount of catalyst is responsible for the sites available for reaction. A proper quantity of catalyst may ensure effective and suitable pyrolysis process. The present work starts with solid phase reactions where there relatively greater quantity of the catalyst is required for maximum contact. However later on the system involves solid liquid and gaseous reactions where the excessive quantities of the catalyst may result charring and more complex reactions. That is why in this case of catalytic pyrolysis of spirogyra, the relative quantity of cement catalyst was optimized by heating the mixture of powdered spirogyra with cement at 400 °C for 1 h. Each of the tests was carried out in triplicate and the average values of the products are shown in Table 3. In this case, the progress of reaction was determined by noting a

Table 1 Effects of temperature on the products of pyrolysis of spirogyra

Temp. (°C)	% Liquid	% Gases	% Residue	Density (kg/m ³)
150	8.00 ± 0.90	2.00 ± 0.86	90.00 ± 0.98	1.21
200	15.00 ± 0.90	7.00 ± 0.98	78.00 ± 0.67	1.22
250	16.00 ± 0.90	12.00 ± 1.32	72.00 ± 0.86	1.26
300	17.00 ± 0.96	13.00 ± 1.36	70.00 ± 0.54	1.28
350	18.00 ± 0.88	14.00 ± 1.35	68.00 ± 0.54	1.29
400	18.00 ± 0.89	17.00 ± 1.36	65.00 ± 0.55	1.29

Table 2 Effect of reaction duration on pyrolysis of spirogyra

Test no.	Time (min)	% Liquid	% Gases	% Residue
1	15	6.50 ± 0.9	3.50 ± 0.89	90.00 ± 0.56
2	30	10.00 ± 0.9	8.00 ± 0.99	82.00 ± 0.56
3	45	12.50 ± 0.9	10.50 ± 1.33	77.00 ± 0.56
4	60	20.00 ± 0.9	14.00 ± 1.36	66.00 ± 0.55
5	75	20.00 ± 0.9	15.00 ± 1.23	65.00 ± 0.56
6	90	20.00 ± 0.9	14.00 ± 1.30	66.00 ± 0.56

number of parameters including the relative amount of products, residue and density of liquid. The main focus was on the amount and nature of the products. Therefore, the optimum amount of catalyst was determined based on the amount of liquid product and its density.

It is observed that cement catalyst has an important role on the yield of reaction and the nature of products of pyrolysis of spirogyra. The density of the liquid product of the pyrolysis increases with the amount of catalyst, and this may be explained on the basis of the capability of the catalyst to trap water. However, based on the mass balance, it is expected that active sites of the cement may decompose or cause reaction on the water contents of the liquid product, and consequently results in the formation of high density product. The reason for such activity may be the attributed to an increase in density of the liquid product. Furthermore, this trend might also be due to origin of water in the pyrolysis of the biomass, which liberates during the combination of hydrogen and hydroxyl radicals produced by decomposition of cellulosic matter. Based on relatively

small quantity of the residue and improved density, 20 wt.% of the catalyst was selected as the optimum for further tests.

Characteristics of the Liquid Pyrolysate

Pyrolysis of spirogyra results in a liquid product which is non-viscous in nature and mainly composed of water having blackish brown color. This liquid contains very small quantity of inseparable oil which floats on the surface as droplets with a sharp irritating smell. It contains some highly volatile components which can cause eye irritation. We attempted to separate the oily and aqueous fractions using separatory funnel but without success. The liquid product decolorizes KMnO₄ solution, which indicates the presence of unsaturated or carbonyl compounds. However, it was not treated with bromine water due to similarity of colors with bromine. The liquid was found non-flammable due to its large quantity of water. Based on literature reports on pyrolysis of biomass, it is believed that this liquid may also contain flammable compounds like methanol [21]. The amount and nature of these flammable compounds may vary according to the nature of biomass and reaction conditions of the pyrolysis.

Distillation of Pyrolysate for Removal of Water Content

About 200 mL of the liquid product of the pyrolysis was prepared with 20 wt.% of the catalyst at the optimum temperature of 400 °C and for the reaction duration of 1 h.

Table 3 Effect of quantity of catalyst on pyrolysis of spirogyra

Catalyst amount (% mass)	Products of pyrolysis (% mass)			Conversion efficiency (%)	Oil density (kg/m ³)
	Liquid	Gases	Residue		
5	16	9	75	25	1.27
10	17	11	72	28	1.28
15	19	12	69	31	1.29
20	20	15	65	35	1.30
25	21	17	62	38	1.31
30	23	15	62	38	1.31

The process was subjected to fractional distillation. The profile of distillation obtained is shown in Table 4. It can be seen from the table that the major fraction of the liquid product comprises water and non-flammable compounds. However, a small quantity of the liquid product (fraction no. 1) contains highly flammable compounds. Reports in the literature indicated that this flammable fraction could be composed of methanol and acetic acid [23]. The second major fraction of this liquid is a brown paste-like material (fraction no. 4). This is the real feed stock for next pyrolysis.

Earlier, we and other authors reported that this fraction was mainly composed of furans, phenols and anhydro sugars [18, 21–25]. Keeping in view of these reports, the liquid product of the pyrolysis of spirogyra was separated into residual paste-like material of the flammable non-viscous liquid, water and a paste-like residue. This residue was believed to be composed of anhydro-sugars and high-boiling-temperature and high-molecular-weight predominantly oxygenated compounds. These fractions were found in the boiling point range of 60–160 °C.

The liquid product of the pyrolysis of spirogyra was separated into water and a paste-like organic fraction by distillation. The presence of large quantity of water makes this liquid non-flammable. The poor flammable characteristic of this pyrolysate may also be attributed by the presence of large amount of the oxygenated compounds. The paste-like organic fraction was improved by the catalytic pyrolysis [26].

Catalytic Pyrolysis of Pyrolysate of Spirogyra

The paste-like material obtained as residue by fractional distillation of the liquid product of the catalytic pyrolysis was converted into highly flammable oil and gas. The pyrolysis was carried out according to the conditions optimized for spirogyra, which was 20 wt.% of catalyst, reaction temperature of 400 °C and reaction duration of 1 h. However, in the present work, the reaction was completed within 15 min, due to extinction of vapors from the reactor. The vapors were condensed into oil, which was collected in the cold trap. The resulting oil was tested for miscibility in water by pouring 10 mL of it into water. It was found that the oil was immiscible and floating on the

Table 5 Catalytic Pyrolysis of dehydrated pyrolysate of the spirogyra

Test no.	% Oil	% Gas	% Residue
1	29.18	20.07	50.75
2	36.24	14.78	48.98
3	35.90	15.57	48.53
Average	33.77	16.80	49.42
SD	3.98	2.85	1.17

water surface. The tests were conducted in triplicate and the results are presented in Table 5.

The apparent difference among the results of these tests is relatively larger, which is mainly due to greatly different density values of the catalyst and the biomass. In line with this, the presence of moisture contents in the biomass may also significantly influence the catalytic activity. The moisture content interacts with the catalyst and changes its activity in an unpredictable way. However, this situation can be overcome by having a strict control over the moisture content and ensuring the biomass miscibility. It has been noticed that both the resulting oil and gas of the pyrolysis process were highly combustible. In the second and third tests, during which a miniature burner was connected to the pyrolysis line after the reaction, the fuels were observed to burn consistently and continuously throughout the process. The gas was never analyzed for its chemical constituents, but was believed to comprise (1) carbon monoxide as suggested by the bluish ting in the flame, (2) hydrogen due to the presence of unsaturated compounds in the product as shown in Table 6 and (3) some hydrocarbon gases based on the literature reports [21, 23]. Based on the average of the three tests, the conversion efficiency of this process was calculated to be 50.6 %.

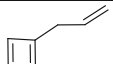
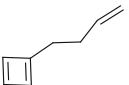
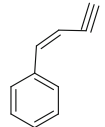
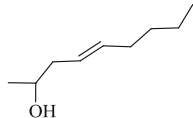
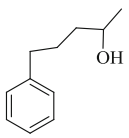
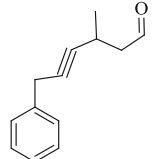
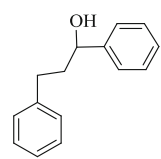
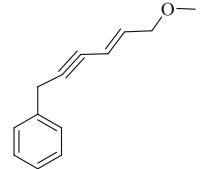
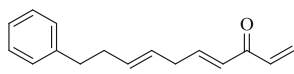
Analysis of Oil using GC-FID and GC-MS

The liquid and oily pyrolysates obtained from the last pyrolysis of the process were analyzed using 6890 N gas chromatograph (Agilent), which was equipped with a flame ionization detector (FID). An HP.5 column was used for separation of oil into its components. The fused capillary column was 30 m in length and 0.32 mm in internal diameter; it also had a 0.25 µm film thickness of stationary phase of polydimethylsiloxane. Nitrogen was

Table 4 Fractional distillation of the pyrolysate of spirogyra

Fraction no.	Temp. (°C)	% Weight	Properties
1	60–80	4.0	Flammable
2	100–120	67.0	Non-flammable liquid
3	130–160	1.5	Non-flammable liquid
4	Decomposable residue	27.5	Paste-like brown material

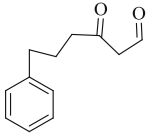
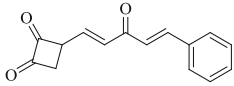
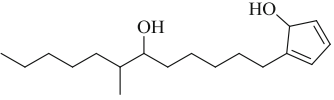
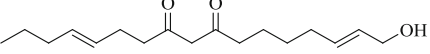
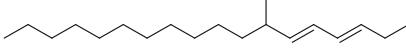
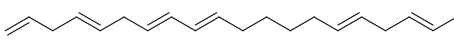
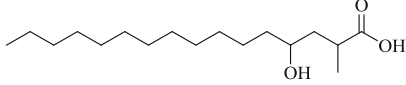
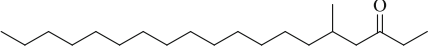
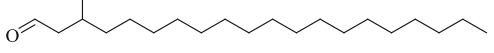
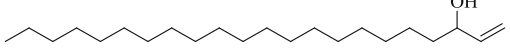
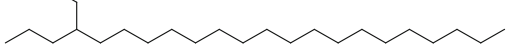
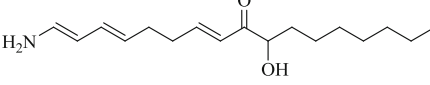
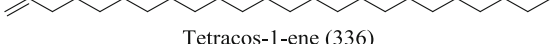
Table 6 Chemical composition of the oil obtained by the second step catalytic pyrolysis of *Spirogyra*

Scan no.	Structure and name (molecular weight)	Relative concentration (%)
30	 1-Allylcyclobuta-1,3-diene (92)	8.3085
45	 1-(But-3-en-1-yl)cyclobuta-1,3-diene 106)	2.1390
53	 (Z)-But-1-en-3-yn-1-ylbenzene (128)	1.0218
96	 (E)-Non-4-en-2-ol (142)	1.2304
388	 5-Phenylpentan-2-ol (164)	10.9770
433	 3-Methyl-6-phenylhex-4-ynal (186)	7.1070
439	 1,3-Diphenylpropan-1-ol (212)	7.5666
467	 (E)-(6-Methoxyhex-4-en-2-yn-1-yl)benzene (186)	6.7610
479		4.5727
502		4.4066
	(4E,7E)-10-Phenyldeca-1,4,7-trien-3-one (226)	

used as carrier gas for this analysis at a flow rate of 1.8 mL/min. The mode of injection was splitless and 1 μ L of the oil was injected to gas chromatograph. For the first

2 min the column was maintained at 75 $^{\circ}$ C and thereafter was heated up to 250 $^{\circ}$ C at the rate of 5 $^{\circ}$ C/minute. The injector temperature was kept at 260 $^{\circ}$ C. As for the GC–

Table 6 continued

Scan no.	Structure and name (molecular weight)	Relative concentration (%)
514	 3-Oxo-6-phenylhexanal (190)	9.8300
563	 3-((1E,4E)-3-Oxo-5-phenylpenta-1,4-dien-1-yl)cyclobutane-1,2-dione (240)	4.932
576	266	4.9600
637	 2-(6-Hydroxy-7-methyldodecyl)cyclopenta-2,4-dienol (280)	5.1320
623	 (2E,13E)-1-Hydroxyheptadeca-2,13-diene-8,10-dione (280)	10.4198
644	 (3E,5E)-7-Methyloctadeca-3,5-diene (264)	1.2198
651	 (4E,7E,9E,15E,18E)-Icosa-1,4,7,9,15,18-hexaene (270)	0.7342
673	 4-Hydroxy-2-methylhexadecanoic acid (286)	0.8555
701	 5-Methylnonadecan-3-one (296)	0.7130
727	 3-Methylcosanal (310)	0.8100
751	 Docos-1-en-3-ol (324)	0.2181
773	 4-Ethylacosane (338)	1.1310
805	 (1E,3E,7E)-1-Amino-10-hydroxyheptadeca-1,3,7-trien-9-one (279)	3.6120
824	 Tetracos-1-ene (336)	1.3420

MS analysis, a 600H Jeol GC–MS was used. All the test conditions and column mentioned for the GC–FID analysis were used in the GC–MS, except for the use of helium gas as carrier and mode of injection, which was

divided for GC–MS operation. It was observed that the nature of products remained the same but the relative amount of the liquid and gaseous fractions changed with temperature.

Analysis of Oil Formed in Secondary Pyrolysis

The oil obtained from pyrolysis of the pyrolysate of spirgyra was analyzed using GC, FID and GC–MS. It was observed that the oil contained large quantities of the less-oxygenated compounds and some hydrocarbons. The resulting chemical compositions of the oil obtained by the second step catalytic pyrolysis are presented in Table 6. As shown in the Table, the oil is composed of 11.47 % cyclic and unsaturated hydrocarbons, 3.30 % long chain olefins, 1.13 % alkanes, 19.99 % alcohols of various types, 3.61 % amino-group-containing compound, 11.35 % aldehydes, 0.86 % carboxylic acid and the rest (48.29 %) are mixed functional group oxygenated compounds. It can be seen from the results that the oily liquid contains very small quantity of acid, which is a highly oxygenated compound that may cause corrosion. Previous reports indicated that significant fractions of pyrolysates of biomass and cellulosic materials could contain large quantities of carboxylic acids [1, 2]. The most abundant oxygenated compounds are alcohols which are less oxygenated in comparison to furans and furaldehydes that are found in the pyrolysates of biomass and paper [25]. Almost all the oxygenated compounds of the resulting oil in this work are of long chain with relatively less percentage of oxygen. Careful calculation of the oxygen contents of the oil indicates that it is composed of 8.379 % oxygen, unlike the oxygen contents of bio-oil obtained by non-catalytic process (in the range of 30–40 %) [14, 27]. The oxygen content was calculated on the basis of the relative wt.% of the compound and mass of the oxygen in one molecule. This indicates that the catalyst plays an important role in lowering the oxygen content [28].

Conclusions

Spirogyra was successfully converted into oil and fuel gas by two step catalytic pyrolysis. The chosen catalyst, cement, was found effective in improving the yield and nature of the product of pyrolysis. The first step was intended to increase the density of the bio-oil and carbon contents, and to fix the gaseous organic compounds into liquid material. It was observed that the density of bio-oil successfully increased by using the cement catalyst. The second stage pyrolysis was aimed to obtain oil free of water and highly oxygenated compounds. It was concluded that the cement catalyst is helpful in reducing the carboxylic acid to a negligible amount and producing significantly higher quantities of the hydrocarbons. Further the catalyst was found effective enough to reduce the total oxygen contents from 30–40 % to 8.379 %. The results imply that the spirogyra, which is a fast growing

biomass, has the potential to be used in large scale production of oil and fuel gases using a low cost and renewable catalyst.

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